Chemistry 3A – Spring 2026 RETAKE VERSION

Sections 19320-19321

Midterm Examination #3 8 May 2026

Indicate **BY DIRECTLY CIRCLING** CLEARLY on the exam page *one* and *only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

1. Which law states that **pressure** and **volume** are inversely related, that is $P_1V_1 = P_2V_2$?
   (a) Boyle's Law (b) Charles' Law (c) Gay-Lussac's Law (d) Avogadro's Law
   LS8 slides 11-14

2. Convert a pressure of 1.50 atm to millimeters of mercury (mmHg)
   (a) 507 mmHg (b) 760 mmHg **(c) 1,140 mmHg** (d) 1,500 mmHg

LS8 slides 6-8. Involves use of a CONVERSION FACTOR, something addressed the whole semester

$$1.50 \text{ atm} \times \frac{760 \text{ mmHg}}{1 \text{ atm}} = 1{,}140 \text{ mmHg}$$

3. In a redox reaction, the species that ***loses electrons*** is said to be:
   (a) reduced (b) precipitated **(c) oxidized** (d) unchanged

LS10 slides 48-52

4. A chemical reaction where two or more simple reactants combine to form a single, more complex product (e.g., A + B → AB) is what type of reaction?
   (a) decomposition (b) single displacement (SD)
   (c) double displacement (DD) **(d) combination (synthesis) reaction**

LS10 slides 15-16

5. What is molarity of aqueous solution in which 0.50 moles KCl dissolved to final 500.0 mL volume?
   (a) 0.25 M (b) 0.50 M **(c) 1.0 M** (d) 2.0 M

LS9 slides 34-37. Given moles directly, and a volume in mL (must convert to L)

$$\frac{0.50 \text{ mol KCl}}{500.0 \text{ mL}} \times \frac{1000 \text{ mL}}{1 \text{ L}} = \frac{1.0 \text{ mol KCl}}{\text{L}} = 1.0 \text{ M}$$

Two significant digits in result

6. If you take 10.0 mL of a 2.5 M stock solution of NaCl and dilute it with water to a final volume of 50.0 mL, what is the final concentration (C_2)?
   (a) 0.10 M **(b) 0.50 M** (c) 1.25 M (d) 5.0 M

LS9 slides 42-43. This is a dilution operation: $C_1V_1 = C_2V_2$ The "C_2" was a hint.

$$C2 = \frac{\text{C1V1}}{\text{V1}} = \frac{(2.5\ M)(10.0\ mL)}{50.0 \text{ mL}} = 0.50 \text{ M}$$

Two significant digits in result because of 2.5 M value

7. A solution is prepared by mixing 25.0 g of solute with 75.0 g of solvent. What is the percent by mass (m/m%) of the solute?

(a) 25.0% (b) 33.3% (c) 75.0% (d) 100.0%

LS9, slides 31-32.

$$\frac{\text{solute mass}}{\text{solution mass}} \times 100\% = \frac{\text{solute mass}}{\text{solute mass} + \text{solvent mass}} \times 100\% = \frac{25.0\ \text{g}}{25.0 + 75.0\ \text{g}} \times 100\% = 25.0\%$$

8. Calculate the **pressure** of n = 2.00 mol of gas at T = 273 K, V = 44.8 L. (R = 0.0821 L·atm/[mol·K])

(a) 1.00 atm (b) 2.00 atm (c) 0.500 atm (d) 22.4 atm

LS8, slides 22-27. You have a value for n, T, and V, AND YOU ARE GIVEN a value for R, which you should recognize as the ideal gas law constant! You are asked for P. All that adds up to using *PV = nRT*, one of the most important equations in chemistry and often given as a problem to solve. Do algebra to get P on one side of equation, and substitute values to solve:

$$P = \frac{nRT}{V} = \frac{(2.00\ \text{mol})(0.0821\ \text{L atm/[mol K]})(273\ \text{K})}{44.8\ \text{L}} = 1.00\ \text{atm}$$

Three significant digits in result appear to be correct.

I want you to note that 22.4 L is the volume of 1 mole of ideal gas at STP; STP has pressure = 1 atm and temperature = 0°C. Here we have 2 moles of gas, and the temperature 273 K = 0°C. The volume is 44.8, which is 2 x 22.4!

9. Convert a **pressure** of 760 mmHg to atmospheres (atm).

(a) 0.500 atm **(b) 1.00 atm** (c) 7.60 atm (d) 76.0 atm

This is another conversion factor use in a calculation like question 2. The conversion factor gets flipped here.

$$760\ \text{mmHg} \times \frac{1\ \text{atm}}{750\ \text{mmHg}} = 1.00\ \text{atm}$$

The significant digits might be wrong here, although you are asked to choose the best response, which you would do. The value "760" would have a clear three significant digits if the value had been given as "760." (note the decimal point)

10. Which of the following is a ***combustion*** reaction?

(a) $CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$ (b) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$

(c) $HCl + NaOH \rightarrow NaCl + H_2O$ (d) $CaCO_3 \rightarrow CaO + CO_2$

LS10, slides 20-21 (with video). Choice (b) is a precipitation talked about in slides 15,16 and can be understood in the Green Sheet. Choice (c) is an acid-base neutralization in slide 39. Choice (d) is a decomposition, gas-evolving (CO2) reaction discussed in slide 18.

11. Which of the following is the correct oxidation number of ***hydrogen*** in H_2O molecule?

(a) 0 (b) -1 **(c) +1** (d) +2

LS10, slide 56, one of the "Oxidation Number Rules"

12. A substance that completely dissociates into ions when dissolved in an aqueous solution and conducts electricity strongly is classified as a:

(a) non-electrolyte (b) weak electrolyte **(c) strong electrolyte** (d) nonpolar molecule

LS9, slide 11

13. What is the oxidation number (ON) for a neutral atom of Cl in its elemental state, Cl_2?

(a) -1 **(b) 0** (c) +1 (d) +7

LS10, slide 55

14. The strongest type of ***intermolecular force*** that exists between two molecules of H_2O (water) is:
(a) dispersion forces (London) (b) dipole-dipole forces
(c) ion-dipole forces **(d) hydrogen bonding**

LS9, slides 6-10 in addition to lectures before this one!

15. How many grams of KOH are required to prepare 1.00 L of a 0.500 M solution?
(molar mass of KOH = 56.11 g/mol)
(a) 5.61 g **(b) 28.1 g** (c) 56.1 g (d) 112.2 g

LS9, slides

$$1.00\ \text{L} \times \frac{0.500\ \text{mol KOH}}{1\ \text{L}} \times \frac{56.11\ \text{g KOH}}{1\ \text{mol KOH}} = \text{g KOH}$$

Both the volume and concentration are 3 significant digits, and so is the result. Calculator shows 28.055 ➔ human thinking 28.1

16. Which is a TRUE statement of the **Kinetic Molecular Theory** for an *ideal gas*?
(a) gas molecules exert significant attractive forces on each other
(b) the molecules that make up the gas phase must interact with their liquid phase
(c) gas collisions are completely elastic (energy is not lost in collision)
(d) the volume (*V*) of the gas is inversely proportional to the number of moles (*n*)

LS8, slide 9. Choice (a) contradicts the one of the elements of KMT, which is that gas molecules must show no attractive forces (electrostatic, magnetic, etc). Choice (b) is just a distraction, not mentioned in KMT. Choice (d) is another distractor, not at all mentioned with KMT

17. Which of the following compounds is ***insoluble*** (forms precipitate) in water?
(a) KNO_3 (b) Na_2SO_4 **(c) AgCl** (d) LiCl

LS10, slides and your Green Sheet. KNO_3, Na_2SO_4, and LiCl are all very soluble. The lectures mentioned several times where silver halides, particularly AgCl were insoluble.

18. A solution that has not yet reached its upper limit of dissolved solute is described as:
(a) unsaturated (b) supersaturated (c) saturated (d) concentrated

LS10. When talking about solubility limits of solutes, the terms unsaturated, supersaturated, and saturated were discussed. While "concentrated" is a term, it is not used in the context of discussing the solubility of compounds which becomes solutes in the solutions.

19. A gas has initial pressure of 2.00 atm and volume of 5.00 L. If the volume is compressed to 2.50 L at constant temperature, what is the new pressure?
(a) 1.00 atm **(b) 4.00 atm** (c) 5.00 atm (d) 10.0 atm

LS This is a straightforward calculation related to Boyle's Law, where $PV = k$, and so $P_1V_1 =$ P2V2. The questions is to solve for a pressure, usually P2

$$P_2 = \frac{P_1V_1}{V_2} = \frac{(2.00\ \text{atm})(5.00\ \text{L})}{2.50\ \text{L}} = 4.00\ \text{atm}$$

The volume is reduced by half, so it makes sense the pressure doubles.

20. Hydrogen bonding can only occur between molecules where hydrogen is directly bonded to one of three highly electronegative atoms. These atoms are:
(a) S, Cl, Br (b) Cl, Br, I **(c) N, O, F** (d) Na, K, Li

LS10 slide 53, and also goes back to LS6, slide 92

21. In a sugar-water solution, the sugar is the ***solute*** and the water is the ***solvent***. The solvent is defined as the component present in:

(a) the smallest amount (b) the solid phase

(c) the greatest amount (d) the ionic form

LS9, slide 4

22. The concentration unit of ***molarity*** (***M***) is correctly expressed in which of the following units?

(a) mol/kg **(b) mol/L** (c) gl/L (d) kg/mol

LS9, slide 34

23. Which of the following is a spectator ion in the reaction (hint: compare complete ionic to net ionic):

$NaCl\ (aq) + AgNO_3\ (aq) \rightarrow AgCl\ (s) + NaNO_3\ (aq)$?

(a) Na^+ (b) Ag^+ (c) Cl^- (d) AgCl

LS10, slides 43-45

Complete ionic: $Na^+\ (aq) + Cl^-\ (aq) + Ag^+\ (aq) + NO_3^-\ (aq) \rightarrow AgCl\ (s) + Na^+\ (aq) + NO_3^-\ (aq)$

Net ionic: $Cl^-\ (aq) + Ag^+\ (aq) \rightarrow AgCl\ (s)$

There are two spectator ions: Na^+ and $NO3^-$. But only Na^+ was given as one of the choices. Choices (b) and (c) are ions, but are not spectator. Choice (d) is not even an ion, but a solid compound.

24. Which statement best describes the principle of "like dissolves like"?

(a) polar solutes dissolve in polar solvents (b) nonpolar solutes dissolve in nonpolar solvents

(c) ionic solutes dissolve in polar solvents **(d) all of choices (a) to (c)**

LS9, slide 6-10.

25. A precipitate forms and you have to write the chemical reaction equation. How will you indicate the compound that precipitated?

(a) (s) (b) ***(l)*** (c) ***(g)*** (d) ***(aq)***

LS10, slide 6

Circle either TRUE or FALSE whether the statement is true or false.

26. A solution that conducts electricity must contain ions.
True False

LS9 slides 11-12, LS10, slide 75

27. The reaction HCl (*aq*) + NaOH (*aq*) → H_2O (*l*) + NaCl (*aq*) is an acid-base neutralization reaction type.
True False

LS10, slide 39

28. The balancing of a chemical reaction equation involves changing the subscripts of a chemical compound formula, like changing **$CuCl_2$** to **CuCl**
True **False**

LS10, slide 10,11. The coefficients, not anything in the compound formula.

29. The oxidation number of a pure element is always zero.
True False

LS10, slide 55

30. In a chemical reaction equation, compounds to the ***right*** of the **reaction arrow** (yield symbol) are called ***reactants***.
True **False**

LS10, slide 3. The **products** are to the right of the arrow, **reactants** to the left.

31. The oxidation number of any Group 1 metal (like Na or K) in any compound is always +1.
True False

LS10, slide 55

32. The total pressure of a gas mixture is equal to the sum of the partial pressures of all the individual gases in the mixture.
True False

LS8, slides 32-36

33. The solid that forms from a solution during a chemical reaction, often by mixing two solutions, is called a **precipitate**.
True False

LS10, slides 11, 25, 34

34. At STP, the temperature is 0°C and pressure is 1 atm.
True False

LS8, slide 28

35. The symbol ***n*** in the Ideal Gas Law equation (*PV=nRT*) is known as the ***universal gas constant***.
True **False**

LS8, slides 22-23. *R* is the "ideal gas law constant" and also called the "universal gas constant". The value ***n*** in the equation refers to the amount (in units of **moles**) of gas that is present.